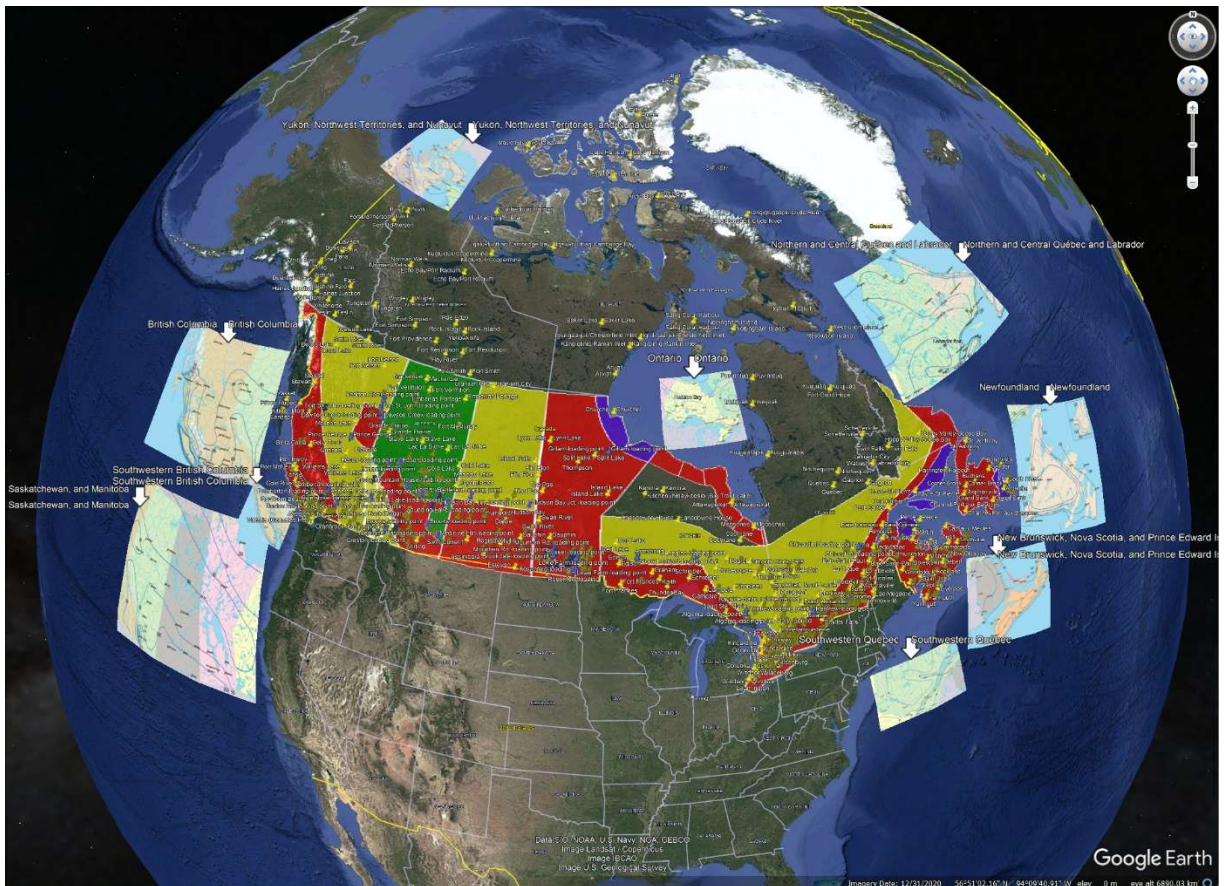


Canada Loading Visualizer for power line design

Include all the weather stations in Canada.



Ver.1.3

Based on CSA C22.3 NO.1-20

Wilson (Haixiang) Huang

2025-11-17

0 Before start

I developed an interactive visual system using Google Earth. This helps all the technologists and Engineers quickly identify their line locations' weather patterns. Instead of jumping back and forth between Google Earth and the Standards.

This Visual system is created based on CSA standard C22.3 NO.1 2020 Version. Which includes all the weather stations information shown in Annex D Table D.2, the Canada loading zones classification shown in Annex C Figure. C.1~5. And the 50-year return period wind speed figures shown in Annex D Figure D.2~10.

The excerpt of the standard is also included in the folder.

Now is Version 1.3

Name: CSA C22.3 NO.1-20_Canada Loading Zones Visualization-2020-Ver.1.3.kmz

1 Disclaimer

The locations of the wind stations shown in this visualization correspond only to their associated cities and do not represent the actual station coordinates. They are provided for reference purposes only.

Similarly, the provincial and national boundaries displayed here are approximate. The CSA standard does not provide precise geospatial boundaries, so the loading-zone outlines in this system are drawn based on the reference points and regional descriptions given by CSA. These boundaries should be interpreted as generalized representations rather than exact administrative borders.

2 Copyright

These materials are intended for internal circulation among current RR Power employees and may otherwise be shared at the discretion of Wilson (Haixiang) Huang.

3 File structure and system description

This KMZ file organizes the CSA C22.3 NO.1:20 loading-related information into three main parts.

- ▼  [CSA C22.3 NO.1-20 Canada Loading Zones Visualization-2020.kmz](#)
Author: Wilson(Haixiang) Huang
Date: 2025/11/17
- ▼  CSA C22.3 NO1:20 Loading Data
 - ▶  CSA Canada Loading Zones
 - ▶  Climatic data for selected Canadian locations
 - ▶  50-year return period wind speed(km/h)

3.1 CSA Canada Loading Zones

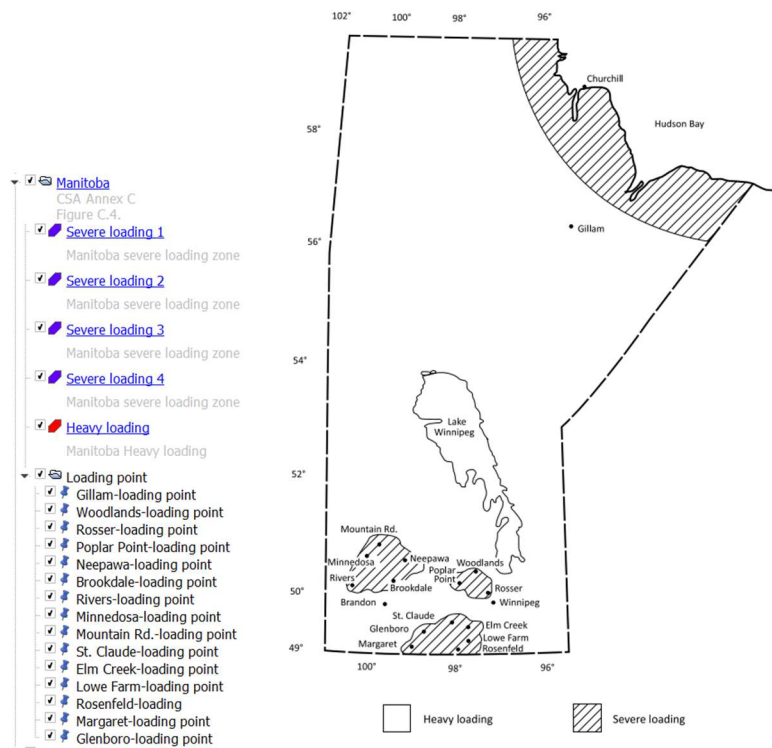
CSA Canada Loading Zones shows the approximate distribution of Medium A, Medium B, Heavy, and Severe loading regions across Canada, based on the regional descriptions provided in the CSA standard. This folder contains the loading-zone visualizations for each Canadian province covered in CSA C22.3 NO.1:20. The provinces are organized individually, and each folder includes the corresponding figure from **CSA Annex C**:

- ▼  CSA C22.3 NO1:20 Loading Data
 - ▼  CSA Canada Loading Zones
 - ▶  [Ontario, Québec, New Brunswick, Nova Scotia](#)
CSA Annex C
Figure C.1.
 - ▶  [Saskatchewan](#)
CSA Annex C
Figure C.1.
 - ▶  [British Columbia](#)
CSA Annex C
Figure C.2.
 - ▶  [Alberta](#)
CSA Annex C
Figure C.3.
 - ▶  [Manitoba](#)
CSA Annex C
Figure C.4.
 - ▶  [Newfoundland](#)
CSA Annex C
Figure C.5.

Inside each province, take Manitoba, for example:

In the Manitoba folder, the **loading-zone polygons** and the **loading-point folder**. The loading zones represent the Heavy and Severe areas shown in CSA Annex C (Figure C.4), while the **loading points (blue pins)** correspond to the reference cities used in the CSA standard to define and divide these zones. These points are simply the CSA reference locations that help indicate where each loading region applies.

Figure C.4
Loading map for Manitoba
 (See Clause 7.2 and Table 30.)

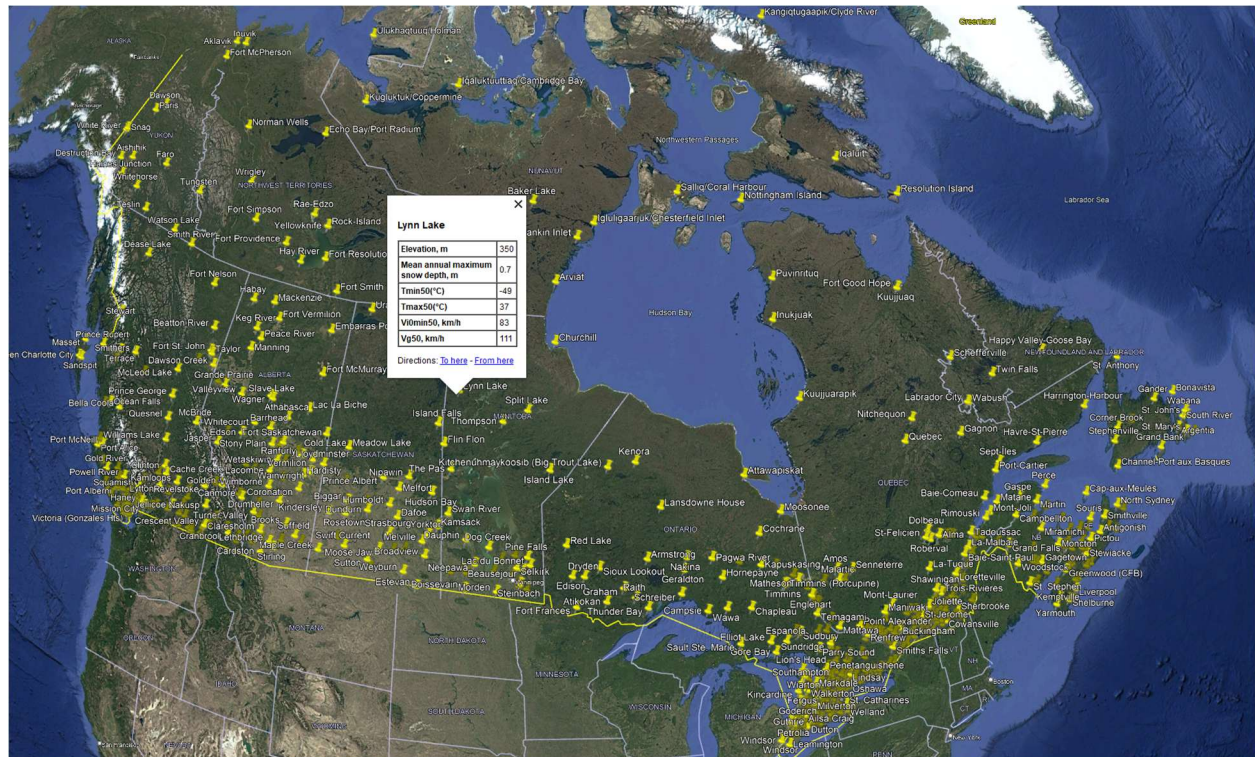


For the loading zone polygon colors:

- Blue ---- Severe
- Red ---- Heavy
- Yellow ---- Medium loading B
- Green ---- Medium loading A
- White ---- Mountain Range for BC

3.2 Climatic Data for Selected Canadian Locations

In this layer, I export all the information from CSA22.2 No.1-20 Table D.2, **shown in yellow pins**, including all the wind stations in Canada. Users can get all the weather values by clicking the point.



3.3 50-year Return Period Wind Speed (km/h)

This layer simply displays the figures from the CSA standard showing the 50-year return period wind speeds. The images are included directly in the system so users can quickly view the reference wind-speed values without needing to open the standard separately.

